Advanced Machine Learning (AML-MDS)
Evaluation rubric for the project, Fall 2026–27

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Abstract

This document is the evaluation rubric for the course project, which is described in the separate *project guidelines*. The three milestones and the final integrative report are graded on the same 10 criteria, each on a 5-level scale, with different weights (Section 3); the proposal and the oral presentation have short rubrics of their own (Sections 5 and 6). The rubric emphasizes theoretical rigor, methodological soundness, and clarity of exposition.

1 Grading scale and interpretation

Each criterion is evaluated using five categorical levels with associated numerical ranges:

- **Excellent** (9.0–10.0): Outstanding work that exceeds expectations
- **Very good** (7.0–8.9): Strong work that meets all expectations with minor imperfections
- **Good** (5.0–6.9): Acceptable work that meets basic expectations but has notable limitations
- **Borderline** (3.0–4.9): Work that barely meets minimum standards with significant deficiencies
- **Poor** (0.0–2.9): Work that fails to meet minimum standards
- **Not delivered**[1]: 0

2 The deliverables and what grades them

Deliverable	Course weight	Graded with
Proposal	5%	Section 5
Milestone 1 — kernel methods	15%	criteria 1–10, milestone weights
Milestone 2 — neural methods	15%	criteria 1–10, milestone weights
Milestone 3 — hybrid and ensemble methods	15%	criteria 1–10, milestone weights
Final integrative report	15%	criteria 1–10, final-report weights
Oral presentation and defence	10%	Section 6
Final examination	*25%*	*not covered here*

3 Weight intervals

Each criterion has a weight interval (percentage of the grade of one milestone or of the final report). The weights used this year are in the last two columns; they stay within the intervals. A milestone weighs the methods of its block and their comparison with what came before; the final report weighs the writing, the conclusions and the whole picture.

[1] UPC regulations state that the final grade must be NP (not shown) when no evaluation has taken place (exams included).

Criterion	Interval (%)	Milestones	Final report
1. Methodological appropriateness and justification	12–18	17	13
2. Theoretical rigor and understanding	12–18	15	15
3. Experimental design and resampling protocol	8–12	11	8
4. Quality and interpretation of results	10–15	11	12
5. Critical analysis and comparison of methods	8–12	12	12
6. Data preprocessing and feature engineering	6–10	8	6
7. Clarity and organization of written report	8–12	8	12
8. Scientific conclusions and insights	8–12	8	12
9. Reproducibility and code quality	6–10	6	6
10. Use of references and acknowledgment of sources	4–8	4	4
Total		**100**	**100**

4 Detailed rubric

4.1 Criterion 1: Methodological appropriateness and justification

Weight: 12–18%

Level	Description
Excellent (9.0–10.0)	Methods chosen are highly appropriate for the problem and course part. Choices are thoroughly justified with clear reasoning connecting problem characteristics, method properties, and theoretical foundations. Demonstrates sophisticated understanding of when and why methods are applicable.
Very good (7.0–8.9)	Methods are appropriate and well-justified. Justifications show good understanding of method properties and problem requirements. Minor gaps in connecting all aspects of the reasoning.
Good (5.0–6.9)	Methods are generally appropriate with basic justification. Some reasoning provided but lacks depth or completeness. May miss some important considerations in method selection.
Borderline (3.0–4.9)	Methods are marginally appropriate or poorly justified. Justifications are superficial or inconsistent. Shows limited understanding of why methods were chosen.
Poor (0.0–2.9)	Methods are inappropriate for the problem or course part, or no justification is provided. Demonstrates fundamental misunderstanding of method applicability.

4.2 Criterion 2: Theoretical rigor and understanding

Weight: 12–18%

Level	Description
Excellent (9.0–10.0)	Demonstrates deep theoretical understanding of all methods used. Mathematical formulations are precise and correct. Assumptions are clearly stated and their implications discussed. Connects theory to practice seamlessly. Provides rigorous analysis of method properties, convergence, or theoretical guarantees where applicable.
Very good (7.0–8.9)	Shows solid theoretical understanding with correct mathematical descriptions. Most assumptions and implications are identified. Minor imprecisions in mathematical notation or theoretical discussion. Good connection between theory and practice.
Good (5.0–6.9)	Demonstrates basic theoretical understanding. Mathematical descriptions are mostly correct but may lack precision. Some assumptions stated but discussion of implications is limited. Adequate but not thorough theoretical treatment.
Borderline (3.0–4.9)	Shows superficial or incomplete theoretical understanding. Mathematical descriptions contain errors or significant imprecisions. Assumptions are unclear or not stated. Weak connection between theory and practice.
Poor (0.0–2.9)	Lacks theoretical understanding. Mathematical formulations are incorrect or absent. No discussion of assumptions or theoretical foundations. Fundamental theoretical errors present.

4.3 Criterion 3: Experimental design and resampling protocol

Weight: 8–12%

Level	Description
Excellent (9.0–10.0)	Experimental design is rigorous and appropriate. Resampling protocol (cross-validation, holdout, etc.) is correctly implemented and well-justified. Statistical significance is properly assessed. Potential sources of bias are identified and addressed. Hyperparameter tuning is done correctly using nested resampling or equivalent.
Very good (7.0–8.9)	Sound experimental design with appropriate resampling protocol. Most aspects of validation are handled correctly. Minor issues in hyperparameter tuning or bias control. Statistical testing is mostly appropriate.
Good (5.0–6.9)	Adequate experimental design with basic resampling protocol. Some aspects may be suboptimal but core methodology is sound. Hyperparameter tuning may have minor flaws. Limited discussion of statistical significance.
Borderline (3.0–4.9)	Experimental design has significant flaws. Resampling protocol is poorly implemented or inappropriate. Hyperparameter tuning is flawed (e.g., using test set). Little attention to statistical validity.
Poor (0.0–2.9)	Experimental design is fundamentally flawed. No proper resampling protocol or serious methodological errors (e.g., data leakage, testing on training data). No consideration of statistical validity.

4.4 Criterion 4: Quality and interpretation of results

Weight: 10–15%

Level	Description
Excellent (9.0–10.0)	Results are presented clearly with appropriate metrics and visualizations. Interpretation is insightful, going beyond surface observations. Uncertainty quantification (confidence intervals, error bars) is provided. Results are contextualized within problem domain and related work. All tables and figures are properly explained.
Very good (7.0–8.9)	Results are well-presented with appropriate metrics. Good interpretation with some insights. Most tables and figures are well-explained. Some uncertainty quantification provided. Minor gaps in contextualization.
Good (5.0–6.9)	Results are presented adequately with basic interpretation. Metrics are appropriate but analysis could be deeper. Tables and figures are present but explanations may be brief. Limited uncertainty quantification.
Borderline (3.0–4.9)	Results are poorly presented or interpretation is superficial. Some tables/figures lack explanation. Metrics may be inappropriate or incomplete. No uncertainty quantification.
Poor (0.0–2.9)	Results are unclear, incomplete, or incorrectly interpreted. Tables and figures are unexplained or misleading. Inappropriate metrics or no proper evaluation.

4.5 Criterion 5: Critical analysis and comparison of methods

Weight: 8–12%

Level	Description
Excellent (9.0–10.0)	Provides thorough, insightful comparison of methods — within the block, and with the baseline and the earlier milestones on the same problem — along multiple dimensions (accuracy, interpretability, computational cost, theoretical properties). Critical analysis identifies strengths, weaknesses, and trade-offs. Discusses why certain methods perform better/worse with theoretical or empirical support. Shows mature scientific judgment.
Very good (7.0–8.9)	Good comparison of methods with reasonable critical analysis. Most relevant dimensions considered. Identifies main strengths and weaknesses. Some theoretical or empirical support for observations.
Good (5.0–6.9)	Basic comparison provided focusing mainly on performance metrics. Limited critical analysis. Some strengths and weaknesses mentioned but discussion lacks depth.
Borderline (3.0–4.9)	Superficial comparison with little critical analysis. Focuses only on final numbers without deeper understanding. Misses important trade-offs or differences between methods.
Poor (0.0–2.9)	No meaningful comparison or critical analysis. Methods are not compared, or comparison is fundamentally flawed. No understanding of method differences.

4.6 Criterion 6: Data preprocessing and feature engineering

Weight: 6–10%

Level	Description
Excellent (9.0–10.0)	Data preprocessing is thorough, well-justified, and properly executed. All issues (missing values, outliers, scaling, encoding) are appropriately handled with clear reasoning. Feature engineering (if applicable) is creative and theoretically motivated. Preprocessing decisions are clearly documented and justified.
Very good (7.0–8.9)	Good preprocessing covering all major issues. Decisions are justified and mostly appropriate. Feature engineering shows good understanding. Minor issues in execution or justification.
Good (5.0–6.9)	Adequate preprocessing addressing basic issues. Some decisions may be suboptimal but overall approach is reasonable. Limited feature engineering or basic transformations only.
Borderline (3.0–4.9)	Preprocessing is incomplete or poorly executed. Some important issues not addressed. Decisions are poorly justified or inappropriate. Little to no feature engineering when needed.
Poor (0.0–2.9)	Minimal or incorrect preprocessing. Major issues (e.g., data leakage in preprocessing) or important data problems ignored. No justification for decisions.

4.7 Criterion 7: Clarity and organization of written report

Weight: 8–12%

Level	Description
Excellent (9.0–10.0)	Report is exceptionally clear, well-organized, and professionally written. Logical flow from introduction to conclusions. Sections are well-balanced. Technical content is explained precisely without unnecessary detail. Writing is concise and free of errors. Figures and tables are well-integrated into text.
Very good (7.0–8.9)	Report is clear and well-organized. Good logical flow with minor structural issues. Writing is generally clear with few errors. Most technical content well-explained. Good integration of figures and tables.
Good (5.0–6.9)	Report is adequately organized but may have structural weaknesses. Writing is understandable but could be clearer or more concise. Some technical content unclear. Figures and tables present but integration could be better.
Borderline (3.0–4.9)	Report organization is weak with poor logical flow. Writing is unclear or verbose. Significant grammar/spelling errors. Technical content difficult to follow. Poor integration of figures and tables.
Poor (0.0–2.9)	Report is poorly organized and difficult to follow. Writing is unclear with numerous errors. Technical content is incomprehensible or missing. Figures and tables are disconnected from text.

4.8 Criterion 8: Scientific conclusions and insights

Weight: 8–12%

Level	Description
Excellent (9.0–10.0)	Conclusions are insightful and well-supported by results. Clearly articulates what was learned and its significance. Identifies limitations honestly and thoughtfully. Suggests meaningful extensions or future work. Reflects deep understanding of the problem and methods. Personal learning is thoughtfully discussed.
Very good (7.0–8.9)	Good conclusions supported by results. Main learnings are clearly stated. Limitations identified. Reasonable suggestions for future work. Shows good understanding and reflection.
Good (5.0–6.9)	Adequate conclusions that summarize main findings. Basic discussion of what was learned. Some limitations mentioned. Limited discussion of extensions or future work.
Borderline (3.0–4.9)	Weak conclusions that barely summarize results. Vague discussion of learning. Limitations not clearly identified or no suggestions for improvements. Little reflection or insight.
Poor (0.0–2.9)	Conclusions missing, trivial, or unsupported by work. No meaningful discussion of learning or limitations. No reflection on the work done.

4.9 Criterion 9: Reproducibility and code quality

Weight: 6–10%

Level	Description
Excellent (9.0–10.0)	Code is well-organized, clearly commented, and follows good programming practices. All results are fully reproducible with clear instructions. Random seeds are set appropriately. Code is efficient and readable. Dependencies are clearly specified. Instructions are complete and easy to follow.
Very good (7.0–8.9)	Code is well-organized and commented. Results are reproducible with minor effort. Good programming practices mostly followed. Instructions are clear and mostly complete. Minor issues with efficiency or documentation.
Good (5.0–6.9)	Code is functional and adequately organized. Results are reproducible but may require some effort. Basic comments provided. Instructions are present but could be clearer. Some code quality issues.
Borderline (3.0–4.9)	Code is poorly organized or documented. Reproducibility is difficult or uncertain. Instructions are incomplete or unclear. Significant code quality issues. Poor programming practices.
Poor (0.0–2.9)	Code is disorganized, undocumented, or non-functional. Results are not reproducible. Instructions are missing or incorrect. Fundamental code quality problems.

4.10 Criterion 10: Use of references and acknowledgment of sources

Weight: 4–8%

Level	Description
Excellent (9.0–10.0)	References are comprehensive, appropriate, and properly cited. All sources (papers, code, data, tools including AI assistants) are acknowledged. Citations are well-integrated into text. Bibliography is properly formatted. Shows engagement with relevant literature. The use of LLMs, if appropriate, is clear and follows recommendations.
Very good (7.0–8.9)	Good use of references with proper citations. Most sources acknowledged. Bibliography mostly well-formatted. Engages with main relevant literature. Minor citation issues. The use of LLMs, if appropriate, follows recommendations with some dubious use.
Good (5.0–6.9)	Adequate references with basic citations. Main sources acknowledged though some may be missing. Bibliography formatting has minor issues. Limited engagement with literature. The use of LLMs, if appropriate, is not totally clear or does not follow recommendations.
Borderline (3.0–4.9)	Insufficient or poorly cited references. Important sources not acknowledged. Bibliography poorly formatted. Minimal engagement with literature. Possible citation issues. The use of LLMs, if appropriate, is insufficient or wrong.
Poor (0.0–2.9)	References missing, inappropriate, or improperly cited. No acknowledgment of sources. No engagement with literature. Serious citation problems or potential plagiarism. The use of LLMs, if appropriate, is absent (this is considered plagiarism and cheating).

5 The proposal

The proposal is graded on the six items the project guidelines ask for, each on the same 0–10 scale, with equal weight. Approval is a separate matter: a proposal can be well written and still be sent back because the problem does not suit the course (too small, too large, no supervised target, data that cannot be shared; for a methodological question, one that cannot be asked of every family of methods in the course). A proposal sent back is resubmitted, and the resubmission is the one graded.

Item	An excellent answer
1. The question	States one precise question, and why its answer matters to someone.
2. The data	Gives source and licence, n, d, the type of every variable, the missing values and the target, all checked on the actual file. For a methodological project, also the generator of the synthetic data: what it varies, over what range, and what is known exactly.
3. The loss	Names the loss and justifies it from the problem; says which predictor it makes optimal.
4. The baseline	Fits a regularized linear or logistic model, says how the penalty was chosen, and reports its result with the protocol of item 5.
5. The protocol	Describes the resampling and the held-out data, and makes clear that nothing is tuned on the held-out part.
6. Expectation and references	Argues what a non-linear method might add on this problem (for a methodological project: how each family of methods might answer the question); cites one or two relevant references; says what the pair expects to learn.

6 The oral presentation and defence

Graded on the same five-level scale, with three criteria.

Criterion	Weight	What is assessed
Presentation	30%	Structure, clarity, and keeping to the time: the story of one problem through the course, not a list of experiments.
Understanding	40%	Answers to questions on the methods used: what they optimize, what they assume, why they behaved as they did on this problem.
Ownership	30%	Each member can justify every choice and explain any part of the code. Code you did not write is code you cannot defend.

Both members are asked questions, and may receive different grades.

7 Notes on application

- Criteria 1–10 apply to the three milestones and to the final report, with the weights of Section 3.
- **Methodological projects** (the fourth way in of the project guidelines) are graded on the same criteria, with these readings. Criterion 3 also covers the design of the synthetic study: what is varied, over what range, with how many replicates, and how the real dataset is used. The theoretical or algorithmic work — a derivation, an analysis of a special case, an implementation, a modification of a method — is assessed under criteria 2 and 8. The empirical part is not optional, so criterion 9 keeps its weight.
- Students should be aware that theoretical rigor (criterion 2) is heavily weighted and applies to every project, whether it starts from a dataset or from a methodological question.
- The instructor may adjust weights within the specified intervals for a project whose nature requires it (for instance, a methodological project); the pair is told in advance.
- Intermediate scores within each level range (e.g., 7.5 for very good) can be considered the most probable grade, but can be altered to reflect nuanced performance.